COHEN-MACAULAY MORPHISMS OF ALGEBRAIC SPACES TESTED ON ETALE CHARTS AND AT POINTS VIA STRICT LOCAL RINGS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6240-6263 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6265-6308, the complete author proof (42 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Cohen-Macaulay morphisms of algebraic spaces, etale charts, and strict local rings; the descent and the pointwise comparison are retained.

Source reference. Original lemma algebra-lemma-quotient-strict-henselization: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6308–6308. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-algebra-lemma-henselization-CM: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6308–6308. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-lemma-CM: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6308–6308. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-flat-at-point-etale-local-rings: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6308–6308. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-properties-lemma-describe-etale-local-ring: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6308–6308. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume the fibres of f are locally Noetherian. The following are equivalent*

(1) *f is Cohen-Macaulay,*
(2) *f is flat and for some surjective étale morphism $V \to Y$ where V is a scheme, the fibres of $X_V \to V$ are Cohen-Macaulay algebraic spaces, and*
(3) *f is flat and for any étale morphism $V \to Y$ where V is a scheme, the fibres of $X_V \to V$ are Cohen-Macaulay algebraic spaces.*

Given $x \in |X|$ with image $y \in |Y|$ the following are equivalent

(a) *f is Cohen-Macaulay at x, and*
(b) *$\mathcal{O}_{Y,\overline{y}} \to \mathcal{O}_{X,\overline{x}}$ is flat and $\mathcal{O}_{X,\overline{x}}/\mathfrak{m}_{\overline{y}}\mathcal{O}_{X,\overline{x}}$ is Cohen-Macaulay.*

Proof. Given an étale morphism $V \to Y$ where V is a scheme choose a scheme U and a surjective étale morphism $U \to X \times_Y V$. Consider the commutative diagram

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

Let $u \in U$ with images $x \in |X|$, $y \in |Y|$, and $v \in V$. Then f is Cohen-Macaulay at x if and only if $U \to V$ is Cohen-Macaulay at u (by definition). Moreover the morphism $U_v \to X_v = (X_V)_v$ is surjective étale. Hence the scheme U_v is Cohen-Macaulay if and only if the algebraic space X_v is Cohen-Macaulay. Thus the equivalence of (1), (2), and (3) follows from the corresponding equivalence for morphisms of schemes, see More on Morphisms, Lemma more-morphisms-lemma-CM by a formal argument.

Proof of equivalence of (a) and (b). The corresponding equivalence for flatness is Morphisms of Spaces, Lemma spaces-morphisms-lemma-flat-at-point-etale-local-rings. Thus we may assume f is flat at x when proving the equivalence. Consider a diagram and x, y, u, v as above. Then $\mathcal{O}_{Y,\overline{y}} \to \mathcal{O}_{X,\overline{x}}$ is equal to the map $\mathcal{O}^{sh}_{V,v} \to \mathcal{O}^{sh}_{U,u}$ on strict henselizations of local rings, see Properties of Spaces, Lemma spaces-properties-lemma-describe-etale-local-ring. Thus we have

$$\mathcal{O}_{X,\overline{x}}/\mathfrak{m}_{\overline{y}}\mathcal{O}_{X,\overline{x}} = (\mathcal{O}_{U,u}/\mathfrak{m}_v\mathcal{O}_{U,u})^{sh}$$

by Algebra, Lemma algebra-lemma-quotient-strict-henselization. Thus we have to show that the Noetherian local ring $\mathcal{O}_{U,u}/\mathfrak{m}_v\mathcal{O}_{U,u}$ is Cohen-Macaulay if and only if its strict henselization is. This is More on Algebra, Lemma more-algebra-lemma-henselization-CM. □